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Inside the black box of food safety: a qualitative study of 'non-compliance' among food businesses

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Abstract

Issue addressed: This paper examines the meaning of food safety among food businesses deemed non-compliant and considers the need for an insider perspective to inform a more nuanced health promotion practice.

Methods: In-depth interviews were conducted with 29 food business operators who had recently been deemed 'non-compliant' through Council inspection.

Results: Paradoxically, these 'non-compliers' revealed a strong belief in the importance of food safety as well as a desire to comply with the regulations as communicated to them by Environmental Health Officers.

Conclusions: The evidence base of food safety is largely informed by the science of food hazards, yet there is a very important need to consider the practical daily application of food safety practices. This requires a more socially nuanced appreciation of food businesses beyond the simple dichotomy of compliant/ non-compliant.

So what? Armed with a deeper understanding of the social context surrounding food safety practice, it is anticipated that a more balanced, collaborative mode of food safety health promotion could develop, which could add to the current model of regulation.

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Introduction

In Australia, Environmental Health Officers (EHOs) employed by local government monitor food safety practices in the food industry and ensure compliance with food safety regulations based on national standards and state legislation. The discourse of food safety is thus dominated by the language of compliance. Within this discourse, compliance is positioned as the desired outcome emanating from the practice of regulation. As Henson and Heasman have observed though, the processes that take place between the regulation and the compliance/ non-compliance outcome are rarely given any research attention, leaving the food business as essentially a 'black box'.[1] Thus, there is little attention given to the nature of the relationship between food businesses and EHOs who must walk a fine line between enforcement and support.[2]

Food safety is *prima facie* an important area for health promotion research in Australia, given the 5.4 mil cases of food-borne illness annually, costing approximately A$1.2 bil.[3] Yet, in reality, food safety attracts meagre health promotion attention with, for example, just a handful of papers published in the *Health Promotion Journal of Australia* concerned with this issue.[4,5,6] This limited interest has been largely confined to maternal and child health[4] and often considered as a corollary of nutrition rather than as a discrete area of focussed food safety inquiry.[5,6] This unfortunate disconnect between food safety and environmental health and health promotion has been observed elsewhere in the world[7,8] and would appear to be an artefact of the gaze produced by the 'regulation' and 'enforcement' that has become the signature discourse of food safety. Within this discourse, the evidence base is generally confined to the science of food hazards and food safety education as a means of improving compliance. Although preventing and reducing contamination is indeed central to food safety, there is no consideration here of the social or behavioural context of food safety.[9]

In Australia, although food safety regulation appears to garner a high level of community trust,[10] it may not be meeting the needs of some food businesses that may wish to comply but lack the time, money or expertise to do so. Our interest here is not to suggest that regulation does not have an important role, rather it is to suggest that other aspects of food safety practice are worthy of more concerted attention. We acknowledge that EHOs working across Australia do indeed adopt practices which are far more sophisticated than that of 'the food police' and that good regulation requires

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a thoughtful 'craft' rather than simplistic 'enforcement'.[11] However there is surprisingly little evidence to guide this area of practice.

Missing is an insider perspective of food safety from the point of view of the food businesses. The language of regulation and compliance is a top-down discourse in which experts set the standards. In contrast, collaboration involves working together, and therefore must include all perspectives within a partnership. Regulation of food businesses, and in particular small food businesses, where there are limited capacities to respond to regulatory regimes, requires understanding their everyday experience of 'doing' food safety. In particular there is a critical need to understand the situations of those businesses that have been deemed 'non-compliant' in a more contextually rich manner. In this paper, we present some of the findings of a recent research project founded on a collaboration between several local government partners in south-east Queensland, Environmental Health Australia (Qld) Inc. (the professional association for EHOs) and the Australian Centre of Excellence for Local Government.[12] This project qualitatively investigated the meaning of food safety and food safety regulation among a sample of food businesses deemed 'non-compliant'.

Method

Underpinning the methodological stance of this research was the concern for the everyday meanings and practices of food businesses in relation to food safety. Conceptually, we drew inspiration from the anthropological premise that it is important to understand health within everyday contexts of life.[13] With this in mind, we do not assume that the knowledge and behaviours of food businesses are necessarily 'misunderstandings' of expert knowledge, but instead we acknowledge that they are founded on their own local logics and rationality.[13] In contrast to the positivism of food safety science, our anthropological focus here is informed by a non-positivist epistemological stance in which we are interested in the meanings of food safety practice rather than their correctness or truthfulness. All participants in this study were food business operators geographically located in southeast Queensland and had received a recent non-compliance food safety notification (excluding administrative or minor structural non-compliance) after a local government EHO inspection. We recruited from non-compliant food businesses in order to investigate both the challenge of compliance and the different rationalities of food safety practice. This was based, conceptually, on other similar areas of health promotion where disjuncture has been identified between the rationalities of the health promotion message and local-level rationalities in the community.[14–17]

To construct our sample of non-compliers, and in order to meet the confidentiality requirements of the Queensland *Food Act 2006*, details of food businesses receiving at least one non-compliance notification during their last council inspection were obtained through either the *Right to Information Act 2009* or the *Information Privacy Act 2009*. Non-compliances related to issues that would affect food safety, such as food handling and storage and hygiene and maintenance (e.g. storage at unsafe temperatures, inadequate cleaning, pest infestations, etc.) rather than administrative non-compliances (e.g. not displaying a current licence certificate). Stratified purposive sampling was used to then select potential participants who represented a cross-section of non-compliant food businesses. The strata used were business type, size, location, level of non-compliance and cultural diversity. As a qualitative inquiry we chose this sampling strategy not for any intended claim of population representivity, rather we were keen to understand a range of food business contexts, including, for example, small and large, city, suburban and regional, and social diversity in business operators. All participation was voluntary and if a business declined to participate, a similar business in the area was approached. The research design was given ethical approval by the Queensland University of Technology Human Research Ethics Committee (approval number 1300000043).

In-depth interviews

In-depth interviews were conducted with 29 food business owners or managers using a semistructured interview guide that focussed on the 'insider perspective' of food safety. There was no checking up on whether businesses were continuing to be non-compliant or not, rather the focus was on their everyday explanatory frameworks of doing food safety and their local level challenges meeting food safety regulations. Interviews were loosely structured to ensure they remained relevant to the research, but also sufficiently open to enable the insider perspective to emerge.[18] Interviews were conducted at the food businesses at times convenient to the interviewees and varied in length from 45 to 90 min. A major goal of these interviews was to develop an understanding of the local, everyday context of food safety, hence conducting the interviews *in situ* was important to assist in framing the 'insider perspective'. Interviews were recorded then transcribed. The transcriptions were shared across the research team and emergent themes were identified and then used to code the data.

Findings

'We know we could lose our business': food safety in a business context

A consistently strong theme in the interviews was the importance of food safety for the livelihoods of the food business operators. This is somewhat ironic, as the group by definition were 'non-compliers' in some regard. Nevertheless, all participants articulated the importance of food safety, albeit often in terms of the survival of their business:

> For a small business it would mean everything. The chef is the owner and he takes it as seriously as does the rest of the kitchen. (Female restaurant manager)

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> This is a small town – word gets around. You risk everything by cheating. (Female café owner/operator)

For larger food businesses, the implications were more institutional in character but nevertheless very high on the business agenda:

> Massive implications from head office – reports, action plans and personally held responsible. (Male supermarket manager)

For some smaller family-run food businesses, food safety was seen as a heavy responsibility directly related to the success of the business with a wide set of social and economic influences.

> I don't know what I would do; this is the only thing we know how to do. We also support family in [the] home country and I also support my children. I have four children. Without this business it would not only mean a risk to my immediate family but to my family back home. (Female café owner/operator)

The desire to comply was evident too in how people spoke about their experience of regulation:

> It's okay, we are probably due for an inspection soon, and it's always stressful. We just have to do what they tell us to do. (Male take-away franchise manager)

'Hot food hot – cold food cold': what does food safety mean?

In contrast to the preceding theme, meanings of food safety were highly inconsistent across our interviewees. Thus, although there was a generalised view of the importance of food safety, this quickly broke down into a diverse array of very specific food safety issues:

> Hot food hot – cold food cold. We ensure that our customers get what they order every time. (Male take-away franchise manager)

> The number one priority is taking care of personal hygiene. (Female bakery owner/operator)

> Covering up food and using fresh vegetables. (Female juice bar manager)

The specificity in these sorts of responses suggests to us that for many food businesses, food safety is particularised without an overarching schema to promote deeper understanding. In essence, food safety meant only that which was directly relevant to a particular circumstance, as explained by the bakery operator below:

> Everything you need to know to keep the fresh food, where to keep your supplies, everything to run a bakery. I would not know what it meant for another business, for example a sushi shop. It would be all different. (Male cafe/bakery owner/operator)

Broader, overarching meanings of food safety were more apparent in our interviews with larger food business operators, whose food safety practices were more likely to be standardised with routine procedures from a head office:

> It means providing good healthy food, proper storage, serving the food properly, avoiding ill health, using fresh product. We use a similar recipe to the franchise company but sometimes we modify it. (Male franchise restaurant owner/operator)

> *[It's about] making sure no one gets food poisoning – we just go through a routine, once you get it, everything follows. (Male franchise restaurant manager)*

'I make sure the temperature control [is] 60 degrees': the culture of rules

It is interesting that in a group of food businesses officially deemed non-compliant, that we got very little sense of a resistant culture of non-compliance. Counter-intuitively we found a regular emphasis on compliance with a particular emphasis on following rules in response to the often narrow definitions of food safety outlined in the preceding section:

> At the end of the day we don't have a job if we don't have food safety. We have to fit by the guidelines that are brought out to us. That means everything to us. We are striving to be 100% compliant with everything we do. (Male take-away owner/operator)

> I run the shop the way they say it to me – I make sure the temperature control [is] 60 degrees. (Female take-away owner/operator)

> Following the regulations of Council and ensuring good quality service to customers. (Male club manager)

If food safety means keeping hot food hot then, the procedural interface for achieving compliance is making sure the temperature control is set at 60°C. The specificity of the procedure helps to transform food safety into a concrete practice or series of practices, but at the same time conceptually reduces food safety to a mechanical process in danger of becoming perfunctory:

> We have regular audits – 760 questions – it's very in-depth. Head Office check that actual routines are in place. As part of the audit they will check the records for the past 14 days to make sure temperatures have been recorded. If you miss one day, that is an issue. (Male supermarket manager)

Several interviewees were from culturally and linguistically diverse (CALD) backgrounds, running small family food businesses and could not help but compare their experience of food safety 'here' to in their home countries, leading to the observation and challenge of 'strict' rules:

> Much stricter here than in China. More things that we have to do to keep businesses running like licences. (Male take-away manager)

> There is a very high standard here for hygiene compared with other countries. In Vietnam, people don't get sick from food over there like they do here (Female take-away owner/operator)

Sometimes CALD food business operators found that EHOs were not familiar with their food and therefore sometimes gave uninformed

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advice. One CALD participant explained that the EHO did not understand how a traditional food should be prepared.

> I had to demonstrate the issues to them to ensure the Council officers understood the issues. Once they did and they knew what I was talking about and together we were able to develop a solution. (Female, take-away owner/operator)

This instance of negotiation and collaboration clearly goes beyond the top-down regulatory discourse of food safety. Yet in other ways these same small businesses were often reminded that being compliant with regulations was the key to their food safety 'success':

> My daughter is doing the food safety training online – I haven't done the program because I don't understand a lot of English so I sent her instead. You have to do it, if someone doesn't have a licence you can get into trouble from Council. I don't understand this, she doesn't work in the business with me, but they said that someone had to have a licence so I had no choice but to send her instead. (Female food store owner/ operator)

The need to follow instructions was a common feature of describing food safety practice:

> The more you do helps you keep up a good record. . . The way I see it is if you do it they shut up and leave you alone. (Female club manager)

> We have not ever got into trouble yet. I still have trouble sometimes, it's hard but we still have to get there. . . If Council comes and tell us to change something or do something we do, so I know that we are doing the right things they tell us to do. (Female supermarket manager)

> If I have wrong anything, Council will tell me that I can't do this. (Female take-away owner/operator)

> They will help me check. They will say 'this I can use, this I can't use it'. Up to them to tell me what I can and can't do. (Female take-away owner/operator)

Perceived petty rules or personal preferences were, however, considered inappropriate and came closest to a discourse of wilful non-compliance:

> The time when it can become difficult is when you are dealing with the personal opinion of the EHOs. That annoys me. We are happy to be compliant with the laws, but not to have to bend to different officers' opinions. There are some grey areas there that relate to the personal opinion of the safety officer. (Female café owner/operator)

> There is trivial stuff – like one time they were looking in the office and not in the kitchen. They were looking at the cabling for the computers, not the things that we thought they should have been looking at. (Male franchise take-away manager)

We note that the perception of petty regulation can reflect inadequate communication and education. For example, in the quote above it may have been that the EHO was looking for pests in infrequently cleaned warm areas, but without explanation such an action could be interpreted as redundant.

Discussion

Given the paucity of food safety research concerned with the insider experience of food business operators and workers, the scope of our findings here could never be considered conclusive. However, in the act of shedding light on the 'black box' we do suggest some important implications of our work.

First, it is important to acknowledge that in our sample of non-compliers the value of food safety was not disputed. There is no suggestion that at an intrinsic level the importance of safe food supply is questioned by our interviewees. However, problems begin to emerge from the transformation of a generalised value into the cohesive framework of food safety practice. Here, the emphasis on rule-following reflects the long history of regulation as the dominant public health approach to food safety. Ironically, many of the 'non-compliers' in our study saw themselves as very much 'compliers'. From an outsiders' perspective the experiences, as told by our participants of their receiving and responding to the instructions from EHOs provides evidence of non-compliance, yet from an insiders' perspective, these same 'non-compliers' saw the experiences as evidence of their compliance. This reactive compliance founded on responding to non-compliance notices rather than proactive engagement with the regulations is not what was intended by the regulators, but is nevertheless a rational response to a rule-based system and compounded by often limited capacities to engage more fully.

Furthermore, there is clear rationality to conceptualising food safety around the markers most often arising from a particular food business context. This is a highly rational response to a system which emphasises a specific subset of food hazards relevant to a particular set of circumstances. Unfortunately the consequence is a reduction in the scope of food safety to a minimalist construct amenable to a minimalist response, such as ensuring a correct temperature.

Other principles of health promotion practice beyond regulation are needed here. An obvious starting point is the nature of the relationship between the EHO and the food business. Food businesses appreciated EHOs who were supportive and communicated well. This has obvious resonance with health promotion principles of creating supportive environments and developing interpersonal skills (see for example Keleher[19] for an overview of health promotion principles). Further, the need for more contextual approaches to food safety that acknowledge the social and cultural diversity of small family-run businesses[20] clearly needs more attention. This broader view of food safety also aligns with calls for more ecological approaches to food safety interventions and research.[21]

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Conclusion

Food safety involves much more than the science of food hazards, since the management of these hazards must at some point bump up against the messy lived realities of workers in food businesses. Their local knowledge and circumstances deserve our understanding, since it is at this intersection that regulations are interpreted and food safety practices are made concrete. Understanding of this interface ought to inform the communication of food safety as much as the science of food hazards. Yet the very significance of food safety as a public health priority has delivered it to a space where regulation has become something of a catch-all for food safety practice. We suggest that shining a brighter light on the black-box of food businesses could bring a more nuanced insider perspective to the evidence base and might provoke a more diverse array of health promotion approaches to this vitally important area of health.

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